Status	Finished
Started	Sunday, 4 October 2026, 8:53 PM
Completed	Sunday, 4 October 2026, 9:00 PM
Duration	6 mins 31 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Heap sort
- b. Merge sort
- c. Selection sort
- d. Quick sort

The correct answer is:
Merge sort

Question **2**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By using the bubble-up method
- b. By sorting data in a single pass
- c. By comparing elements sequentially
- d. By breaking the input into smaller parts and merging them

The correct answer is: By breaking the input into smaller parts and merging them

Question **3**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Bubble Sort
- b. Merge Sort
- c. Linear Search
- d. Quick Sort

The correct answer is: Merge Sort

Question **4**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Searching for data in a list
- b. Arranging data in a particular format
- c. Deleting data from a list
- d. Inserting data into a list

The correct answer is: Arranging data in a particular format

Question **5**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Searching
- b. Sorting
- c. Merging
- d. Rearranging

The correct answer is: Sorting

Question **6**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sorted()
- b. arrange()
- c. order()
- d. sort()

The correct answer is: sort()

Question **7**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It complicates the analysis process
- b. It allows for quicker identification of duplicates
- c. It makes the analysis slower
- d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **8**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 40,80,90,150,200,300,400,470,70,120
- c. 80,150,200,470,40,90,300,400,70,120
- d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **9**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are combined to form a single sorted list
- b. They are discarded
- c. They are split again into smaller sublists
- d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **10**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It only works on small datasets
- c. It simplifies the sorting process by using only one pass
- d. It avoids the need for recursion

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **11**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Linear Search
- b. Binary Search
- c. Merge Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **12**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Comparing each element with the others
b. Sorting the entire list sequentially
c. Dividing the input into smaller subproblems
d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **13**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes it easier to select items based on their relationship to the rest of the items
b. It makes the data unsorted
c. It slows down the process
d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **14**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort does not guarantee sorted order
b. Bubble Sort cannot handle duplicate elements
c. Bubble Sort is difficult to implement
d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **15**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It is less efficient than custom sorting algorithms
- b. It sorts data out of the box efficiently
- c. It requires external libraries
- d. It only works with integer arrays

The correct answer is: It sorts data out of the box efficiently